

Dynamic 1:k Matching Analysis of Preventive Mastectomy Effectiveness in Breast Cancer

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Motivation and background:

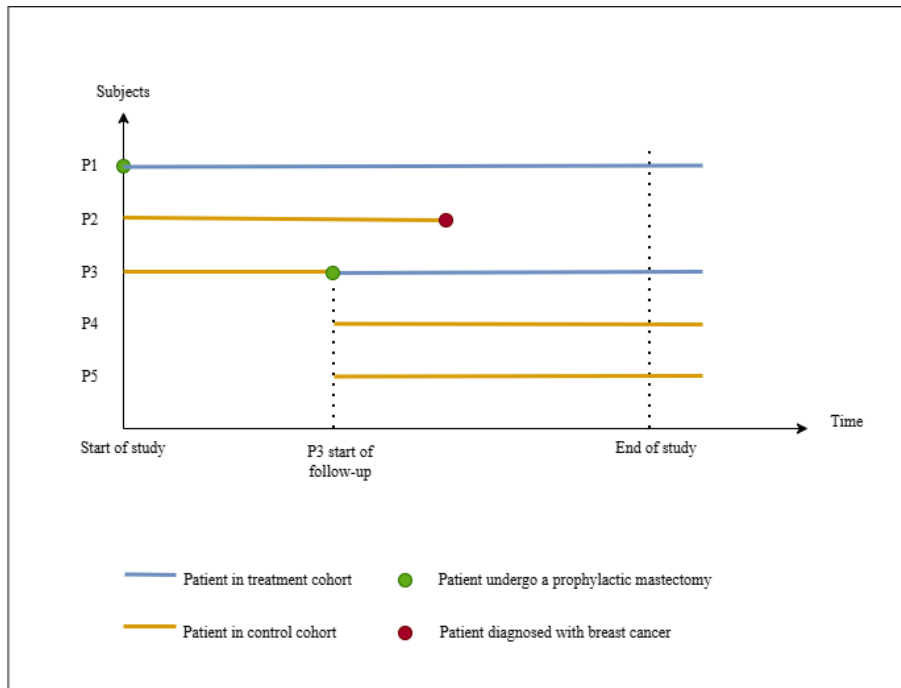
Breast cancer is the most common malignancy among women in Israel and Western countries, with a lifetime incidence of approximately one in eight Israeli women.[1] While the majority of cases (85-90%) are non-hereditary, certain genetic mutations are associated with significantly elevated risk. The BRCA1 and BRCA2 mutations, particularly prevalent among Ashkenazi, Ethiopian, and Iraqi populations, account for 50-80% of hereditary breast cancer cases and substantially increase lifetime disease susceptibility. Although complete prevention of breast cancer remains elusive, effective risk-reduction interventions exist, e.g., Prophylactic mastectomy, which is a surgical removal of breast tissue in high-risk individuals.[2,3]

This study will evaluate the long-term effectiveness of prophylactic mastectomy in high-risk patients using statistical matching. Matching is a technique that estimates treatment effects by pairing treated patients with comparable untreated individuals based on observable related characteristics, thereby constructing a comparison group.[4]

This analysis adapts the methodology of Dagan et al.[5] from Covid-19 vaccination effectiveness to mastectomy effectiveness, while implementing 1:k matching rather than 1:1 matching. Specifically, each treated patient is matched with multiple control patients, an approach anticipated to yield lower variance in the effect estimates. The matching is performed dynamically, allowing cohort composition to change throughout the follow-up period. At each assessment point, control group patients may undergo prophylactic mastectomy and consequently become ineligible as controls. Such patients are transferred to the treatment cohort and matched with k new control patients.[5]

This dynamic matching approach presents a methodological challenge: determining the appropriate follow-up endpoint when multiple matches exist per treated patient. Specifically, the analysis must address whether the follow-up period ends when one matched control undergoes mastectomy, or whether a minimum number of remaining matches is required.

Figure 1: Dynamic 1:2 matching demonstration



Problem definition:

While statistical matching methods exist for static cohorts with multiple controls and for dynamic 1:1 matching, the combination of these approaches, dynamic 1:k matching, lacks established methodology in the literature. This creates several unresolved challenges.

The primary methodological gap concerns follow-up endpoint definition. When a treated patient is matched with k controls and one or more controls subsequently receive treatment, it remains unclear whether follow-up should terminate immediately, continue until a minimum number of matches remain, or follow another rule.

Additionally, appropriate estimators for this dynamic context must be determined. Traditional estimators for static cohorts may introduce bias when cohort composition changes over time, particularly with censored data. In this dynamic framework, censoring occurs when a control patient undergoes a mastectomy and transitions to the treatment group. Standard methods may not appropriately account for this dynamic matching process and may lead to incorrect conclusions about treatment effectiveness.

This project addresses these gaps by developing a comprehensive 1:k dynamic matching framework applicable to evaluating prophylactic mastectomy effectiveness.

Research stages:

1. **Literature review of estimation methods:** Examine existing estimators for simplified scenarios, including static cohorts with complete data and static cohorts with censored data.
 2. **Performance metric specification:** Establish metrics for evaluating estimator quality to guide subsequent development.
 3. **Estimator development:** Propose estimators tailored to dynamic cohorts with censored data.
 4. **Covariate Selection:** Identify and select the covariates to best describe patient matching.
 5. **Simulation and estimator evaluation:** Apply the proposed estimators to both synthesized data and real data from Clalit and Leumit health maintenance organizations. Assess estimator performance using the established metrics.
- Note:** The work with synthesized data will be conducted during the research project since the researcher doesn't have direct access to the real sensitive data. The final evaluation will be conducted on the real data with the help of the supervisor.

Possible extension: Derive theoretical properties of the proposed estimators

Literature Review:

This research examines statistical matching with multiple controls in censored data environments with dynamic cohorts. The literature review draws upon two use cases exemplifying key characteristics of the target environment, extracting methodologically sound approaches applicable to this study. The first is Dagan et al. (2021), which presents 1:1 matching with censored data and dynamic cohorts in assessing Covid-19 vaccination effectiveness [5]. The second is Yoo et al. (2025), demonstrating 1:4 matching with censored data examining the relationship between Parkinson's disease (PD) and benign paroxysmal positional vertigo (BPPV) [10]. Two critical decisions for successful research implementation are identified: matching paradigm selection and analysis method choice. This review evaluates approaches employed in both use cases and justifies selections appropriate for the current research.

The matching paradigm aims to approximate randomized controlled trial (RCT) conditions, making it essential for valid treatment-control comparisons [7]. When implementing multiple control matching, the bias-variance trade-off requires careful consideration. While increased control numbers reduce variance, additional controls may introduce bias, as they are necessarily more dissimilar from the treated patient than the closest match [7].

Yoo et al. employed 1:4 Fixed Matching, assigning exactly four controls per treated patient [10]. Generally, 1:k Fixed Matching assigns precisely k controls per treated patient. This approach typically utilizes optimal matching algorithms that minimize the total distance between treated and control participants across the dataset [6]. Distance is conventionally measured as the absolute difference in propensity scores (or their logits), representing each participant's probability of receiving treatment [6]. Dagan et al.'s 1:1 approach represents a special case of Fixed Matching where $k=1$. Due to its dynamic nature, matching occurred daily for newly treated participants, precluding global optimization in favor of a greedy matching algorithm. This sequential algorithm selects the most similar available control for each treated participant, often requiring similarity within a defined threshold (caliper). While Dagan et al. did not implement a caliper, it represents common practice in the literature [5,6,7]. An alternative not employed in these use cases is Varying Ratio Matching, which permits different control numbers per treated participant. Maximum control numbers may be specified but are not mandatory. Literature favors Varying Ratio Matching for minimizing bias, as Fixed Matching may necessitate dissimilar controls when the control pool is limited [7]. Varying Ratio Matching typically employs the greedy algorithm described above [7]. This research initially proposed Fixed Matching. However, given the advantages noted in the literature, Varying Ratio Matching warrants consideration or comparative evaluation. Regarding matching algorithms, Dagan et al.'s periodic greedy matching approach appears well-suited for dynamic cohort scenarios. For Fixed Matching, optimal algorithms are recommended, with the initial matching at the beginning of the trial employing optimal matching, followed by periodic greedy matching for subsequent additions. For Varying Ratio Matching, greedy algorithms are appropriate throughout.

Analysis method selection is crucial for accurate effect estimation [8]. Dagan et al. employed the Kaplan-Meier survival analysis estimator, appropriate for their 1:1 design [5]. However, Kaplan-Meier assumes observation independence, which holds for 1:1 matching but fails with multiple controls, as controls are selected based on covariate similarity to treated participants, creating dependence. Yoo et al. employed the Stratified Cox Proportional Hazards Model as an alternative [10]. This Cox regression variant estimates treatment effects on time-to-event outcomes for grouped or matched data. The Stratified Cox model offers two key advantages for this research. First, it accommodates varying baseline hazards between treatment-control sets, appropriate given matching based on propensity score distance naturally produces varying hazards across sets. The baseline hazard is defined as a nonnegative function of time, however, it factors out in the partial likelihood formula, eliminating the need for its explicit calculation [9]. Second, it handles censored data by accounting for varying set sizes and follow-up periods. These

characteristics also suit dynamic environments where follow-up periods naturally vary between sets [7,8,10]. An additional implementation advantage is that the Stratified Cox Proportional Hazards Model accommodates varying set sizes, making it suitable for both Fixed Matching and Varying Ratio Matching approaches. Despite being the most appropriate method identified for this research context, the Stratified Cox Model has limitations that warrant acknowledgment. This approach yields conditional estimates of treatment effects rather than marginal estimates. Whereas randomized controlled trials typically report marginal hazard ratios characterizing population-level effects, conditional hazard ratios describe effects at the individual level. Furthermore, conditional estimates may result in amplified treatment effect magnitudes compared to marginal effects [8].

This review has synthesized matching paradigms and analysis methods from relevant literature and empirical research applicable to the current study's objectives. The methodological framework established provides a robust foundation for conducting rigorous causal inference in the presence of censored data and dynamic cohorts, balancing theoretical optimality with practical implementation considerations.

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